

TAG MOHAMED SALAH HAMAD

Junior AI / Python Backend Engineer

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PROFESSIONAL SUMMARY

Motivated Computer Science graduate with hands-on experience in backend development, system architecture, and machine learning. Skilled in Python, C++, SQL, Django, and TensorFlow. Proven ability to design efficient algorithms, build scalable RESTful APIs, and optimize database systems. Seeking a Junior AI/Backend Engineer role to develop AI-driven software solutions.

EDUCATION

Bachelor of Computer Science

Kafr El-Sheikh University, Faculty of Computers and Information

Graduated at: Sept 2024 | Grade: B+

Relevant Coursework: Data Structures, Algorithms, Database Systems, Machine Learning, Natural Language Processing, Operating Systems

TECHNICAL SKILLS

- Programming Languages:** Python, C++, C, SQL, JavaScript, HTML, CSS
 - Frameworks & Libraries:** Django, TensorFlow, PyTorch, Keras, Scikit-learn, Pandas, NumPy, SciPy
 - Databases:** SQL Server (SSMS), PostgreSQL, MySQL
 - Tools & Technologies:** REST APIs, Git, Docker (Familiar), Linux (Basic)
 - Core Competencies:** Object-Oriented Programming, Data Structures, Algorithm Design, Complexity Analysis, Database Design, System Architecture
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PROJECTS

AI-Powered Sentiment Analysis API

Technologies: Python, Django, TensorFlow, REST APIs

- Built a RESTful API that analyzes customer reviews and returns sentiment scores (positive/neutral/negative).
- Integrated a pre-trained TensorFlow model into a Django backend for real-time predictions.
- Designed a PostgreSQL database to store user queries and analysis history.
- Deployed the API locally using Docker for containerization.

Library Management System

Technologies: C++, SQL Server (SSMS), Data Structures

- Developed a console-based system to manage book inventory, member records, and borrowing history.
- Implemented efficient search and sorting algorithms using AVL trees and hash maps for $O(\log n)$ performance.
- Designed a normalized relational database with SSMS and wrote complex SQL queries for reporting.

Customer Churn Prediction Model

Technologies: Python, Pandas, Scikit-learn, Matplotlib

- Cleaned and preprocessed a customer dataset of 10,000+ records using Pandas and NumPy.
- Built and compared Logistic Regression and Random Forest models to predict customer churn.
- Achieved 85% accuracy and presented insights on key churn drivers.

CERTIFICATIONS

- **Backend Development with Python** – Codezilla (Instructor: Eng Islam Mahfouz) | 2020
- **Python Programming Basics** – ITI Mansoura University | 2020
- **Object-Oriented Programming with Python** – ITI Creative, Mansoura University | 2022
- **Data Structures with C++** – ITI Creative, Mansoura University | 2021
- **Database Mastery (Level 1-2) with SSMS** – Eng Mohamed El-Desouky | 2022
- **Python Backend Developer** – Sololearn | 2022

ADDITIONAL COURSES

- Basics of Programming with C – Eng Osama EL-Zero
- TKinter GUI Development – Eng Mahmoud Ra'ouf

LANGUAGES

- **Arabic:** Native
- **English:** Professional Working Proficiency